

Two 8-K event trades, one discipline: facility disruptions × FIRMS, and biotech readouts × Jev

Gator Quant Hacks 2026 · Trade the 8-K · Strategy A: physical facility disruption 8-Ks screened by NASA FIRMS thermal persistence → defined-risk short event-vol (library 5 cash-secured put, extended to an iron condor). Strategy B: small-cap biotech clinical_trial_results 8-Ks graded by Jev → long/short stock with a protective put / call hedge (library 3 and 1). Both pre-registered; `run_study(start, end)` reproduces each on any window.

45 · 24 · 8 A: disruption 8-Ks · FIRMS-brief in-sample · out-of-sample	11.5% → 5.3% A: implied move vs [realized]; realized beat implied in 13%	+2.42% A: iron condor 3/10 edge over placebo (h=21 + expiry)	1,140 / 333 B: readout 8-Ks, in-sample 2024-25 / out-of-sample 2026	p = 0.38 · -23% B: Jev vs day-0 move · put cost after the news vs before
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1 · Hypothesis and novelty

Both strategies start from the same observation: an 8-K is a scheduled re-pricing, and the options chain states a price for it. The question is whether an **outside signal** says that price is wrong. The two signals, and the two sides of the trade, are different.

- A · Disruptions: the market over-prices what the satellite says is not burning.** A plant fire, outage or shutdown 8-K reads like a tail event, and the 1-month chain prices ~11% moves. NASA FIRMS gives an independent read of the site: if no thermal anomaly persists ≥ 3 days the filing is `brief`; if heat persists it is `persistent`. Claim: on `brief` filings realized paths are quiet, so sell premium with defined risk. The core five structures are stock-linked and cannot harvest a volatility edge; the iron condor can.
- B · Readouts: the text carries the signal, the market has it by the close, and the hedge is cheapest after.** A small-cap biotech is one or two programs, so a readout re-prices the company. Claims: (1) Jev can grade the masked press release on a five-level scale; (2) the market under-reacts by the entry close, so sign-following drifts; (3) implied vol collapses once the result is out, so the protective leg is cheaper after the news. Universe: all 1,499 SIC 2834/2836 filers, dead names included — big pharma filed zero such 8-Ks.

Novelty. Neither signal is in the price feed: one is a satellite label, the other a language model reading the filing. Neither trade is a generic long call — A is short vol with wings, B is a hedged directional book whose hedge timing is itself the finding.

2 · Method

	A · Disruptions × FIRMS	B · Readouts × Jev
Events	Massive 8-K <code>items_text</code> → disruption phrases (fire, explosion, outage, shutdown, weather, force majeure) → geocode → FIRMS → <code>brief</code> / <code>persistent</code> / <code>unknown</code> . Trade <code>brief</code> ; <code>persistent</code> is the control, <code>unknown</code> skipped. 55-name industrials / materials / energy universe.	One per accession, filer matched on CIK. EDGAR acceptance minute sets <code>t_0</code> , the first session at whose close the 8-K was public. Jev's announcement date sets the news session; <code>t_pre</code> is the session before, so the implied move is pre-news.
Signal	FIRMS persistence (<code>PERSIST_DAYS</code> = 3) on the geocoded site, applied to every disruption filing.	Nine typed questions per filing, company and drug codes masked: new readout, five-level outcome → <code>sentiment</code> , endpoint met, safety, phase, partner.
Trade	1m chain, entry at the filing close, 5% OTM wings; synthetic spot from the ATM pair; six structures scored, iron condor 3/10 added.	<code>expected</code> = <code>0.78 × sentiment × IM</code> ; trade when <code>P(readout) ≥ 0.7</code> , <code> sentiment ≥ 0.5</code> and <code>r_0</code> is short of half the expected move. 10%-OTM, 21-45-day hedge.
Controls	Fixed horizons 1...21 + expiry; placebo ordinary days on the same names; bootstrap CIs on the difference; 2025 out-of-sample; 5% premium haircut; sensitivity on OTM, bucket, entry session, <code>PERSIST_DAYS</code> .	Placebo (same names, random sessions, random direction); sign-shuffle; no-Jev momentum baseline; stale readouts; 2026 out-of-sample frozen; 95% bootstrap; 28 × 8 sensitivity grid.

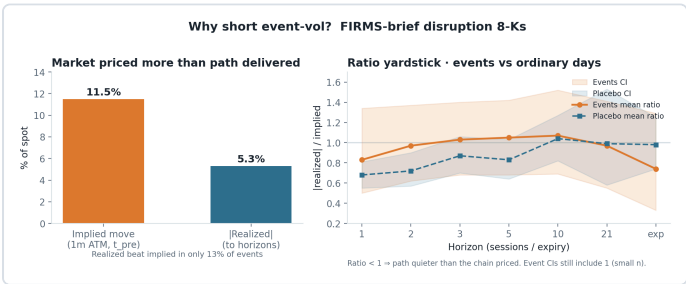


Fig. 1 · A, the yardstick. Median 1m implied move vs realized on FIRMS-brief filings, and the [realized] / implied ratio by horizon against placebo days with bootstrap bands. Below 1 means the chain priced more than the path delivered.

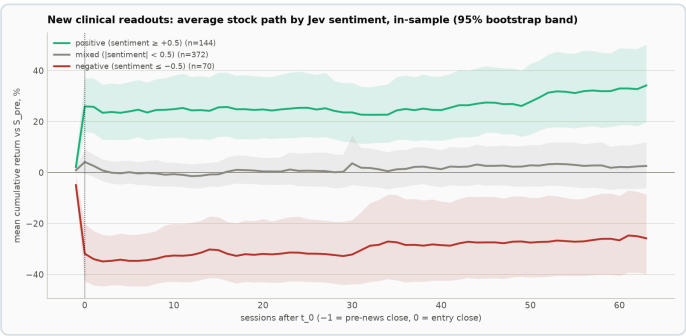


Fig. 2 · B, the signal. Cumulative return from the pre-news close by Jev sentiment, 586 new readouts, 95% bands. The separation is complete at the entry close: the text is right, and it is priced within the session.

3 · Results, with uncertainty

A · Edge in the premium. Realized beat implied in 13% of events. Ranked on $h=21$ + expiry against placebo, **iron condor 3/10** +2.42% ($n=7$) > cash-secured put +1.10% ($n=12$) > long call +0.81%; covered call -3.85%, collar -4.75%, protective put -6.68%. The condor difference from placebo excludes zero at $h=10$ (+0.82%) and expiry (+3.20%); the put and collar losses at $h=21$ also clear the CI. Event ratio CIs still include 1 — the edge is in the P&L of the right structure, not in a single ratio.

Out-of-sample 2025 ($n \approx 5$ priced): the condor keeps its sign mid-horizon (+0.10% $\rightarrow$ +1.76% at $h=21$); hold-to-expiry flips (+1.96% $\rightarrow$ -1.69%), as does the cash-secured put. After a 5% premium haircut each way the condor is -0.78% net at $h=21$ and +0.84% net at expiry, with 22% of events net positive at $h=21$.

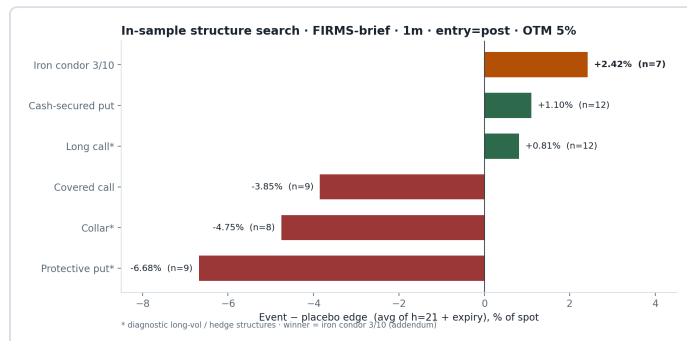


Fig. 3 · A, what wins. Event-minus-placebo edge by structure, FIRMS-brief, 1m, entry = post, 5% OTM. Short defined-risk premium leads; stock-linked and long-vol structures lose.

B · Edge in the classifier and in the hedge, not in the drift. Claim 1 held everywhere: Spearman 0.38 with the day-0 move ($n=586$); 0.75 against 60 blind hand labels, 98% within one level; slope of r_0 on sentiment 0.41 micro-cap vs 0.16 mid+; biotech median $|r_0|$ 14% vs 1.5% for the big-pharma partner. Claim 2 failed: rule minus placebo at 5 sessions -1.1% in-sample, -4.0% out (CI excludes zero at $h=1-5$ out-of-sample); random directions beat the rule 77% / 94% of the time; no gate, size, entry or horizon in the grid turns it positive. Claim 3 held: ATM IV median 117% $\rightarrow$ 100%; a 10%-OTM put costs 8.6% of spot the evening before and 6.3% after (cheaper in 75% of readouts, median -23%). Realized / pre-news straddle median 0.6 at one session, 0.8 at 21.

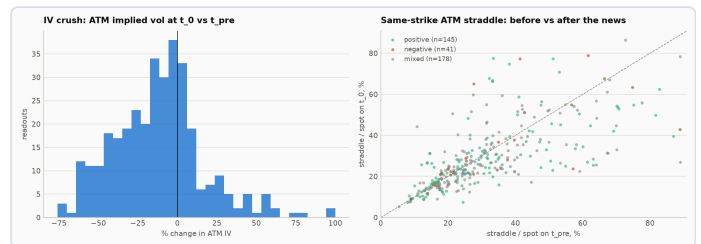


Fig. 4 · B, the hedge is cheaper after. Change in ATM implied vol from t_{pre} to t_0 , and the same-strike straddle as a share of spot before vs after the readout; points below the diagonal are cheaper hedges.

4 · Sealed-window prediction, on record

- A.** The implied-vs-realized gap on brief filings replicates; the mid-horizon condor edge holds weakly or not at all at single-digit n ; the expiry result flips sign again. Any filing the satellite mislabels — a persistent shock read as brief — hurts, with loss bounded by the wings.
- B.** No edge over placebo at any horizon for the directional rule. The day-0 correlation, the size dose-response and the post-event IV drop replicate.
- Both pipelines take (start, end) and nothing else; event caches, classifier version and all fitted constants are frozen.

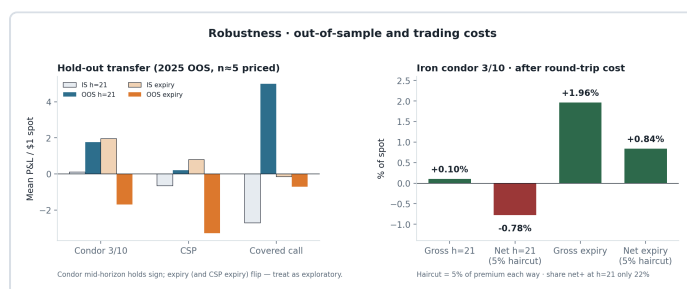


Fig. 5 · A, out-of-sample and costs. In-sample vs 2025 means at $h=21$ and expiry for the condor, cash-secured put and covered call, and the condor's gross vs net P&L after a 5% premium haircut each way. The mid-horizon sign survives; expiry does not.

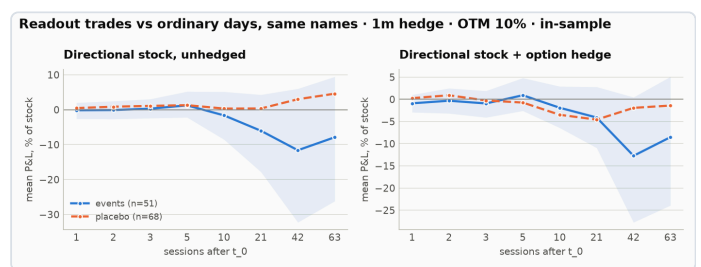


Fig. 6 · B, the directional book vs ordinary days. Mean P&L of readout trades, unhedged and hedged, against placebo sessions on the same names with 95% bands. Nothing separates from placebo before the hedge expires — the prediction we carry into the sealed window.

5 · Trade realism

A. Enter at the filing-session close once the FIRMS label is available; 1m, 5% OTM wings, exit by $h=10-21$ rather than hold to expiry. Single-name OTM volume is thin (median 21 contracts on the legs) and the 5% haircut erases the $h=21$ edge, so size is small and the expiry leg is where the net edge survives. Skip persistent and unknown.

B. Not as a directional 8-K trade — median quoted spread 53% of mid turns +0.9% gross into -3.2% net, and hedge-leg volume caps a trade at about \$548. What is usable: Jev as a same-day classifier (sign right 71% of the time on directional readouts) and the **post-event put** — hold the stock, buy the 10%-OTM 1-month put at the t_0 close, cross the spread, size to the leg's volume.

What a PM takes away. Two independent outside signals, two different places the edge lives. On disruption 8-Ks the satellite says the chain is over-priced, and a defined-risk short-vol structure collects it mid-horizon before costs; on readout 8-Ks the model is right about direction, the market agrees within the session, and the edge is in paying less for the hedge after the print. Both are small- n , cost-sensitive, and stated with the horizon and the failure mode a desk would need.